BTS II(R&S)/I(S)–04.18–0824 Reg. No.

B

B.Tech. Degree II Semester Regular and Supplementary/ I Semester Supplementary Examination April 2018

CE/EE/ME/SE GE 15-1204 A & CS/EC/IT GE 15-1104 B
BASIC ELECTRICAL ENGINEERING
(2015 Scheme)

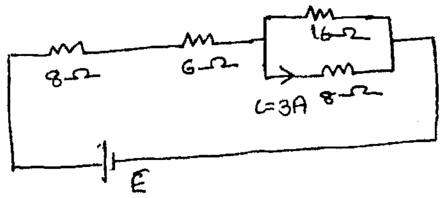
Time : 3 Hours Maximum Marks : 60

PART A
(Answer *ALL* questions)

$(10 \times 2 = 20)$

I. (a) State and Explain superposition theorem.

(b) Find the emf E in the given circuit.



(c) State and explain Faraday's law.

(d) Find the inductance of a toroid of 12.5 cm mean radius, 6.25 cm^2 circular cross-section wound uniformly with 300 turns of wire.

(e) Find the RMS value of sine waveform. Derive necessary steps.

(f) Calculate the power factor of purely capacitive circuit using phasor diagram.

(g) Find an expression for the impedance and p.f of an RL series circuit supplied by an a.c voltage.

(h) In a star connected 3ϕ system, line voltage equal to $\sqrt{3}$ times phase voltage. Explain.

(i) Explain the operation of DC motor.

(j) Draw the circuit diagrams of different types of DC generators.

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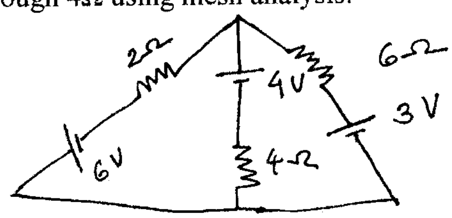
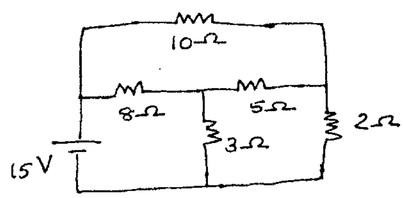
PART B

$(4 \times 10 = 40)$

II. (a) Find the current through 4Ω using mesh analysis. (5)



(b) Determine current through 10Ω resistor. (5)

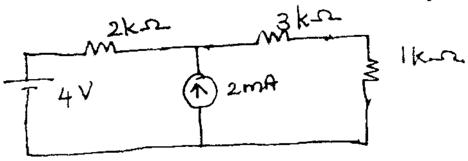


OR

(P.T.O.)

III. (a) State and explain Thevenin's theorem. (4)

(b) Find Norton equivalent circuit for the network faced by 1kΩ resistor. (6)



IV. (a) A mild steel ring having cross-sectional area 500 mm^2 and mean circumference 400 mm has a coil of 200 turns wound uniformly around it, calculate: (6)

(i) Reluctance.

(ii) Current required to produce a flux of 800 μwb.

(b) Explain the working principle of Ammeter. (4)

OR

V. (a) A magnetic field is produced by a coil of 300 turns; which is wound on a closed iron ring. The ring has a cross-section of 40 cm^2 and mean length 150 cm. Permeability of iron is 800. If the current in the coil is 10A, find energy stored in magnetic field. (7)

(b) What is mutually induced emf? Explain. (3)

VI Calculate form factor of sinusoidal current wave form. Derive necessary steps. (10)

OR

VII. (a) Prove that the power drawn by a pure inductor over a complete cycle is zero. (5)

(b) Two circuits, the impedances are given by $z_1 = (20+15j)\Omega$ and $z_2 = (10-4j)\Omega$ are connected in parallel. If the total current supplied is 20A, find power taken by each branch. (5)

VIII. Explain the operation of hydro-electric power plant, with the help of neat diagram. (10)

OR

IX. (a) Explain the working of transformer with neat diagram. (4)

(b) Explain the classification of DC motors. What are the applications? (6)

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